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On-Chip Light-Scattering Enhancement Enabled by a Microlens Array for High-Performance Single-Particle Tracking under Conventional Bright-Field Microscopy

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Scattering-based single-particle tracking (S-SPT) has revolutionized the label-free detection and characterization of nanoscopic objects, offering immense potential for diverse analytical applications. However, the high technical demands placed on optical systems have long impeded its widespread adoption. To address this, an on-chip microlens-based approach that significantly enhances light scattering, thereby reducing the requirements on back-end optical systems, is introduced. Unlike existing field enhancement techniques, which are limited by their highly localized field, this approach leverages enhanced long-range optical fields and complex interactions between nanoparticles and the microlens to achieve an enhancement range ten times greater. This method enables high-performance S-SPT using a conventional bright-field microscope under incoherent light sources with relatively low illumination powers. The approach achieves nanometer localization precision for 60 nm gold nanoparticles in an aqueous medium within a substantial 750 μm^2 field of view at a 200 μs exposure time. This advancement will significantly facilitate the practical application of S-SPT in biosensors and related fields, making it more accessible and versatile for a broad range of research and industrial applications.

1. Introduction

Single-particle tracking (SPT) is a powerful technique that provides crucial insights into the dynamic behavior of individual molecules or particles, shedding light on the intricate mechanisms that govern various physical, chemical, and biological processes at the nanoscale.[1–4] The implementation of SPT heavily relies on light microscopy techniques for the optical detection of single particles, enabling precise localization and trajectory reconstruction. Fluorescence-based light microscopy excels in background suppression, facilitating the detection and localization of labeled species through spectral separation. However, despite its widespread use in biological applications, it is constrained by labeling efficiency and fluorescence saturation, which limit the number of emitted photons and, consequently, impact imaging speed and localization accuracy. In contrast, elastic light scattering-based light microscopy offers a distinct advantage due to its theoretically unlimited photon budget and the universal applicability of the scattering phenomenon.[5–8]

Recent advancements in light-scattering-based SPT (S-SPT) have achieved remarkable sensitivity and spatiotemporal resolution in the label-free detection and tracking of nanoparticles, including single dye molecules, plasmonic nanoparticles, viruses, and small proteins.[9–16] These advancements are driven by innovative illumination, detection, and background suppression techniques. Prominent methods include darkfield imaging, interferometric detection, and surface plasmon resonance microscopy. These capabilities make S-SPT an attractive tool for analytical and clinical diagnostic applications. However, widespread adoption of these technologies is currently hindered by their technical complexity and high costs. They require sophisticated and expensive microscopic systems, highly stable laser sources with intense illumination powers (typically ranging from 1–10 kW cm^{-2}),[8,9,11] and, in some instances, intricate image processing techniques to isolate and maximize the scattering signals.

To address these challenges, on-chip light-scattering enhancement emerges as a viable solution, amplifying the scattering signal before it reaches the optical system. Various near-field techniques have been developed to enhance light–matter interactions through field enhancement, such as surface plasmons or optical antennas.[17,18] Nonetheless, these techniques are unsuitable for

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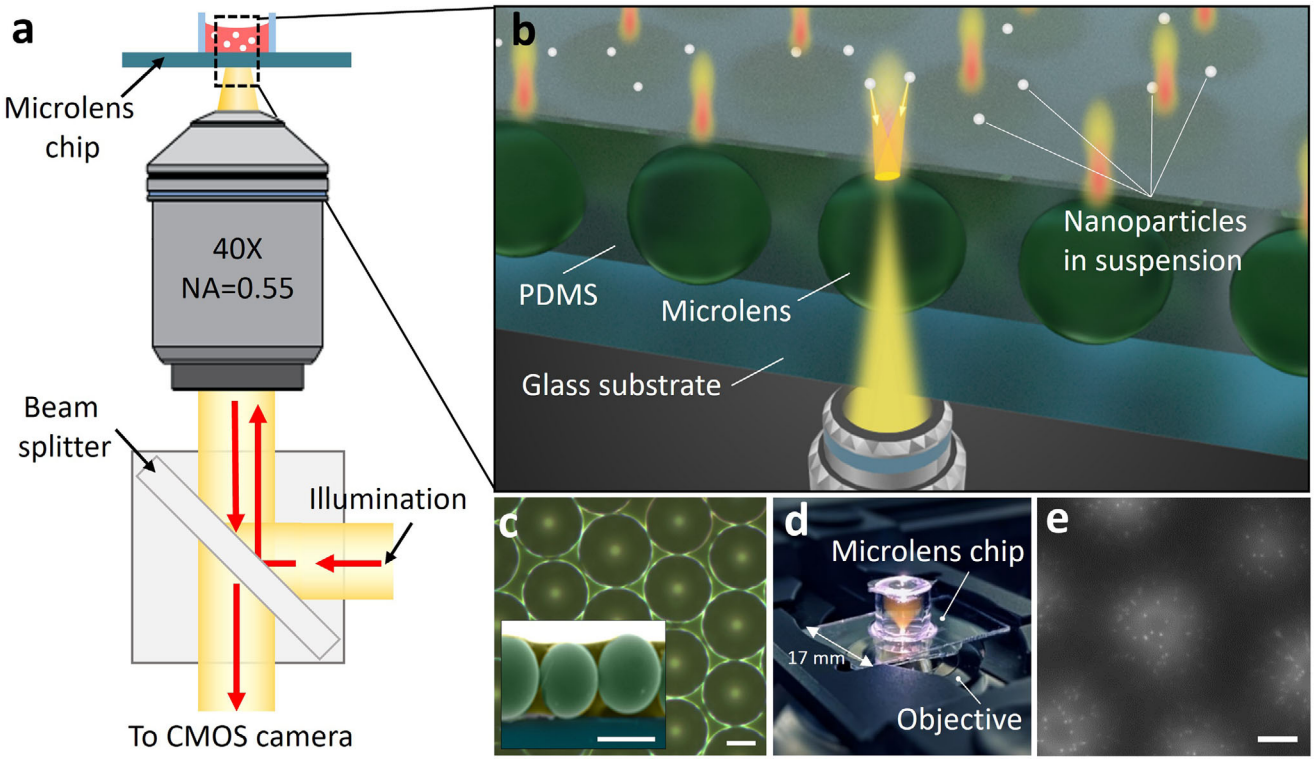

Figure 1. Schematic illustration and experimental setup. a) Schematic of the optical system. The microlens chip is positioned on a conventional inverted optical microscopy system with unpolarized incoherent light source. b) Illustration of the on-chip light-scattering enhancement. The enhanced backscattered light from individual particles is collected by the microlens and then transmitted to the objective. Note that the illustration is not drawn to scale. Specifically, with a 40× objective, the illumination light is deliberately focused on ≈20 microlenses within the field of view. c) Morphology and cross-section (insert image) of the microlens chip. The images are taken by optical microscope and scanning electron microscopy (SEM), respectively. The different colors in the false colored SEM image distinguish the glass substrate, microlenses, and PDMS, respectively. Scale bar = 50 µm. d) Photograph of the microlens chip under working condition. The glass substrate is clamped on the sample stage of the inverted optical microscope. A sample reservoir is made by attaching a glass ring on top of the glass substrate. e) Gray level virtual image of 100 nm AuNPs captured under microlens chip. The bright spots in the image indicate individual AuNPs. Scale bar = 20 µm.

S-SPT due to their highly localized field enhancement, which results in an insufficient range to cover the 3D trajectory of particles within the interrogation time. Additionally, they are inherently costly and demand specific nanostructures and complex near-field configurations.[19] There is a pressing need for a direct and efficient method to achieve on-chip scattering enhancement that transcends the limitations of near-field range.

In this study, we introduce a straightforward and robust on-chip microlens-based strategy for light-scattering enhancement, providing an enhancement range tenfold greater than that of near-field optical techniques. Our approach utilizes dielectric microspheres with a high refractive index (microlens chip), which have been reported as superlenses capable of magnifying nearby objects before projecting them into a conventional microscope's objective lens.[20–24] The microspheres act as microlenses, effectively enhancing the numerical aperture (NA) perceived by the microscope objective, thereby enhancing light-scattering detection capabilities. Our research delves into the unexplored characteristics and applications of these microlenses, revealing their capacity to simultaneously enhance the backscattering of visible light from individual nanoparticles, ranging from near-field to semi-near-field. This enhancement is attributed to the amplified long-range optical fields and complex composite interactions between two closely spaced structures.[25–31] As a result, a sufficient number of detectable photons and a high signal-to-noise ratio (SNR) are achieved, enabling high-performance tracking of individual plasmonic nanoparticles even under a conventional bright-field microscope with incoherent light sources, as illustrated in **Figure** 1a,b. This represents a notable advancement in reducing the technical demands of S-SPT, paving the way for its practical application in biophotonics, biosensors, diagnostics, and beyond.

2. Results

The microlens chip is composed of close packed high refractive index (RI) dielectric microspheres on a glass substrate (Figure 1c) and shares rather simple fabrication process (see Figures S1, S2, Supporting Information). As a demonstration, we used commercially available dielectric microspheres with a RI of 1.92 and diameters of 73.5 µm, selected for their size uniformity (±2.0%). The optical system is based on a conventional inverted optical microscope equipped with a low NA objective lens (NA = 0.55), illuminated by unpolarized incoherent white LED light source (Figure 1a). The microlens chip is positioned above the objective and nanoparticle solution is dropped onto the chip

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(Figure 1d). Each microsphere acts as an individual microlens. In the reflection-based illumination mode, the backscattered light from nanoparticles is collected by the microlenses, captured by objective and finally imaged onto the camera, generating a magnified virtual image in the far field. As a point-like scatterer, individual nanoparticles appear as diffraction patterns on the image plane, known as the point spread function (PSF). We first imaged surface-immobilized gold nanoparticles (AuNPs). Typical image of 100 nm AuNPs observed with the microlens chip is shown in Figure 1e. Each microlens generates a magnified virtual image where the single AuNPs appear as bright spot-like PSFs against the background in the field of view (FOV, $\varphi \approx 25$ µm in real space). Consistent with the characteristics in microsphere-assisted imaging, the central region of the FOV exhibits superior image quality with optimal brightness and minimal distortion. Towards the edges of the FOV, the brightness of their corresponding PSFs decreases, and their circular shape distorts into a spindle-like shape due to the distortion. Aggregated AuNPs can be readily distinguished based on their larger sizes and distinct morphologies, as established by our previously described methodology (see Experimental Section in the Supporting Information). In our study, we define a region of interest (ROI) within the FOV characterized by a distortion rate of less than 1% (see Supporting Information for detailed discussion). This corresponds to an ROI of ≈37 µm^2 ($\varphi \approx 6.9$ µm) for individual microlenses and a total ROI of ∼750 µm^2 for all the microlenses ($n \approx 20$, as indicated in Figure 1c). Accordingly, only the AuNPs within ROI were subjected to analysis. Within this region, the variation in PSF brightness is shown to be less than 9% (Figure S4, Supporting Information). Further details regarding the distortion will be presented later.

To demonstrate the light-scattering enhancement afforded by microlenses, we compare images of surface immobilized 150 nm AuNPs imaged with and without the microlens chip under the same illumination power (**Figure 2a**). This diameter was chosen to ensure a sufficiently intense light-scattering signal while they are observed directly under a 40× objective without microlenses. To compare their photoelectron counts, the gray-level images produced by the CMOS camera are converted and presented in terms of photoelectrons (see Supporting Information).

As expected, AuNPs imaged with the microlens exhibit higher scattering intensity compared to those imaged without the microlens. While bright visible intensity was captured for the AuNPs imaged with the microlens, only inconspicuous intensity was observed for the control samples (Figure 2b). To quantify this enhancement effect, we define enhancement factor δ as $\delta = N_{PE\text{-}\mu lens}/N_{PE}$, where $N_{PE\text{-}\mu lens}$ and N_{PE} denote the integration of net photoelectrons for an individual AuNP imaged with and without the microlens, respectively. The microlens chip significantly boosts the number of photoelectrons by ≈76-fold under the illumination power of 0.7 W cm^2 (Figure 2c). Meanwhile, this considerable boost is not accompanied by a corresponding increase in background noise, resulting in a ≈16-fold enhancement in *SNR*. The observed 76-fold signal increase and 16-fold SNR improvement deviates from expected shot-noise-limited behavior, suggesting that other factors, such as improved system stability, reduced electronic noise, or better isolation from environmental noise, may contribute to the observed SNR enhancement. It is worth noting that the adaptability of this microlens chip allows for its seamless integration with techniques that utilize high NA objectives, thereby significantly boosting their overall performance. In particular, existing state-of-the-art S-SPT techniques that prioritize photon collection can benefit from this enhanced photon acquisition. Additional experiments have shown that the microlens chip enhances the performance of high NA objectives (e.g., 63×, NA1.0, water-immersion objective), yielding a ≈19-fold increase in photoelectrons when imaging single AuNPs, along with a substantial ROI of ≈150 µm^2 (Figure S5, Supporting Information).

We next assess the performance of the microlens chip on smaller AuNPs. Previous studies have revealed that, for AuNPs smaller than 100 nm, their scattering cross-section exhibits a rapid decay with decreasing size.[32,33] A challenge thus arises when imaging small nanoparticles, as a very limited number of scattered photons (proportional to their scattering cross-section) is expected. To address this, we investigated AuNPs with diameters of 100 nm, 80 nm, and 60 nm, respectively. Without the microlens chip, reliable detection of individual AuNPs under a conventional bright-field microscope becomes challenging, making it infeasible to determine their enhancement factor. However, with the microlens chip, these AuNPs can be clearly observed. The dependency of photoelectrons on their sizes imaged with microlenses is shown in Figure 2d. The highly increased abundance of photons allows capturing surface-immobilized single AuNPs with sub-millisecond temporal resolution while maintaining sufficient *SNR* under low-power illumination (Figure 2e). For instance, 60 nm AuNPs can be captured at an exposure time of 200 µs with an illumination power of 1 W cm^{-2} with $SNR > 20$. In comparison to advanced S-SPT techniques, which utilize highly stable laser sources and intense illumination powers ranging from 10 kW cm^{-2} for iSCAT to 2.7 kW cm^{-2} for total internal reflection (TIR) and 1 kW cm^{-2} (equiv. to 10 µW µm^{-2}) for dark-field microscopy,[8,9,11] our study employs incoherent light sources with relatively low illumination power, demonstrating the potential of our method for enabling more energy-efficient and cost-effective SPT.

As the *SNR* scales with the square root of illumination power, it is possible to achieve further enhancements in *SNR*, which can also lead to shorter exposure time when using higher-power sources (Figure S6, Supporting Information). In our study, we primarily focused on 60 nm AuNPs, which are well-established markers in dark-field microscopy and serve as a robust benchmark for comparing detection sensitivities. It is important to note, however, that the detectability of particles in dark-field imaging is influenced not only by their size, but also by the overall performance of the imaging system, including illumination efficiency, collection capabilities, and SNR. Our technique demonstrates the potential to detect even smaller particles. However, achieving reliable detection of smaller particles may require further optimization of imaging parameters and refinement of image processing algorithms. To the best of our knowledge, this is the first time that such small AuNPs can be captured with such high temporal resolution under a bright-field white-light microscope. This enhancement in backscattered photons arises from the enhanced backscattering effect observed in the system, which consists of a nanoparticle and a dielectric microsphere.[25,26,34,35] This effect is attributed to three distinct factors: i) intense optical hotspots focused by microlenses (Figure 2f), ii) improved collection efficiency resulting from the enhanced NA (Figure S7,

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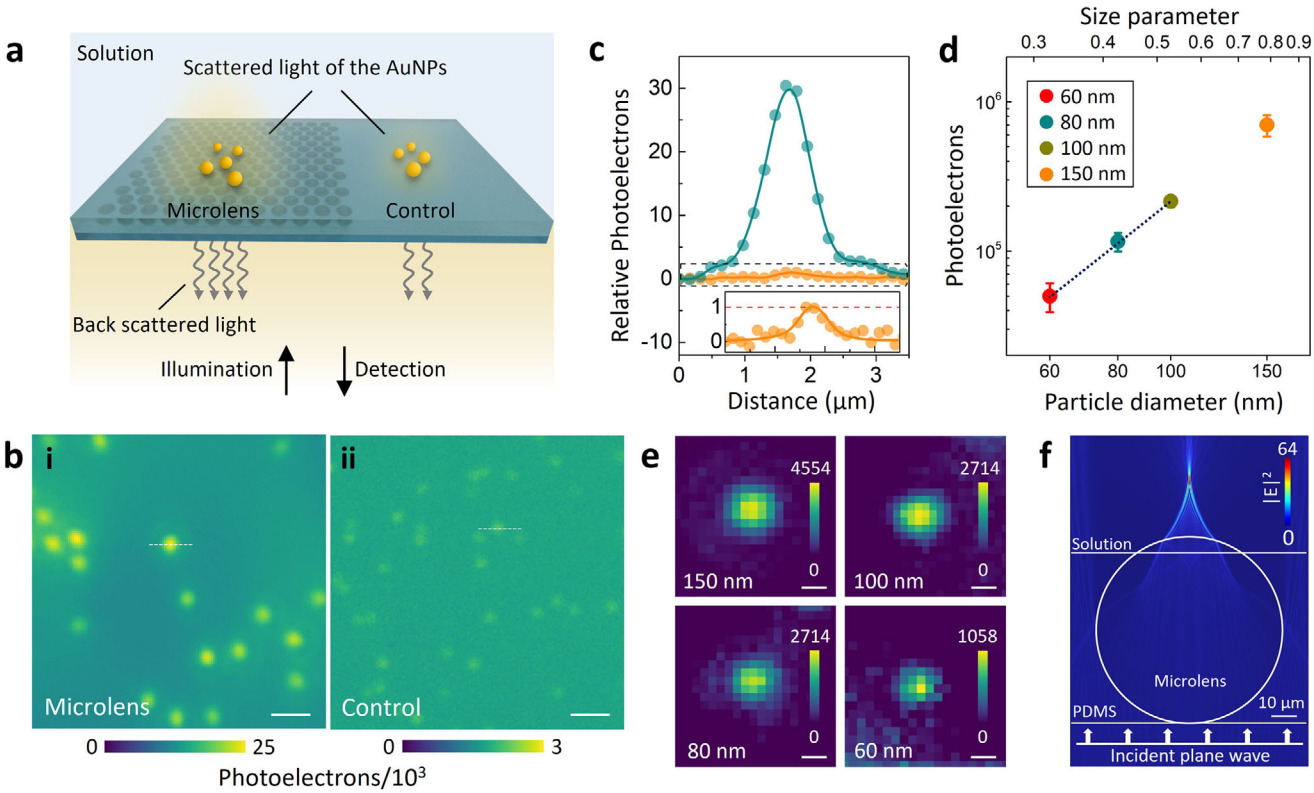

Figure 2. Enhancement effect of the microlens chip on the light-scattering from nanoparticles. a) Schematic illustration of the experimental configuration designed for on-chip light-scattering investigation. Back-scattered light from the AuNPs on the substrate with and without microlenses (control sample) is collected respectively. b) Photoelectron images of 150 nm diameter AuNPs imaged with (i) and without (ii) the use of microlens chip. Photoelectron data were acquired at an exposure time of 1.5 ms with white light illumination of 0.7 W cm^{-2}. Note that for the convenience of visual identification, the scale bars of the photoelectron are different. Scale bar = 2 μm. c) Comparison of their relative numbers of photoelectrons from individual AuNPs indicated by dashed lines in (b). Green and orange dots and lines denote the AuNP imaged with and without the microlens chip, respectively. Note that due to the background variation in (b), the comparison between microlens and control images is depicted by comparing their net photoelectrons. This involves subtracting their respective backgrounds and normalizing by the maximum value of an AuNP imaged without the microlens chip. Inset: the enlarged view of the AuNP imaged without the microlens chip. d) Measured photoelectrons from AuNPs of different sizes on the use of microlens chip. Each point represents an individual AuNP ($n = 5$ particles per size). Data presented as mean ± SD. Photoelectron data were acquired at an exposure time of 1.5 ms with white light illumination of 0.7 W cm^{-2}. The size parameter is defined as $\alpha = \pi d/\lambda$, where d is the diameter of the particle, and λ is the wavelength of the incident radiation ($\lambda = 600$ nm). The dashed line represents the linear fitting of the three values, with a slope of 2.87. e) Relative photoelectron images of AuNPs of different diameter (150, 100, 80, and 60 nm). Photoelectron data were acquired at an exposure time of 200 μs with white light illumination of 1 W cm^{-2}. The photoelectron value in each image is obtained by subtracting its corresponding background value. Scale bar = 50 nm. f) Finite element method simulation of the electric field ($|E|^2$) distribution generated by the microlens. The diameter of the microlens is 73.5 μm and the wavelength of the illumination light is set as 600 nm. The simulation shows that the incident light is converged by the microlens within its FOV at its shadow-side. See also Figure S7 (Supporting Information) for further details.

Supporting Information), and iii) the mutual interactions between AuNPs and microlenses (detailed discussion in Supporting Information).

After the immobilized AuNPs are imaged with high temporal resolution, we extract their positions and investigate the achievable localization precision. Localization precision refers to the accuracy with which the positions of individual nanoparticles can be determined from experimental data. In a shot noise limited system, the localization precision is inversely proportional to the square root of the photon number scattered by the nanoparticle.[9] Consequently, having a surplus of collected photons can lead not only to increases in temporal resolution, but also in spatial resolution. As previously noted, distortion, particularly pincushion distortion, occurs in a microsphere-assisted imaging system and impacts localization precision by altering the lateral magnification of the microlens.[36,37] Besides, the lateral magnification of nanoparticles is also influenced by their vertical position along the z-axis within the depth of field (DOF). We assess the influence of pincushion distortion and DOF on localization precision based on geometric optics (detailed discussion and calculations can be found in Supporting Information).[38] The results show that pincushion distortion exhibits relatively small impacts at the center of FOV. For instance, in the region where $\varphi = 6.9$ μm within the FOV, the distortion rate is less than 1%, resulting in position errors in both the x and y directions ($\delta_{x,y}$) of less than 1% (**Figure 3b**; Figure S3, Supporting Information). As for the DOF, its influence is also minor, with an impact of $\delta_z < 1.1\%$ (Figure S3, Supporting Information). The region within the FOV where $\varphi = 6.9$ μm (≈37 μm^2) is defined as the ROI where the overall position error δ_r is less than 2%. Accordingly, only the

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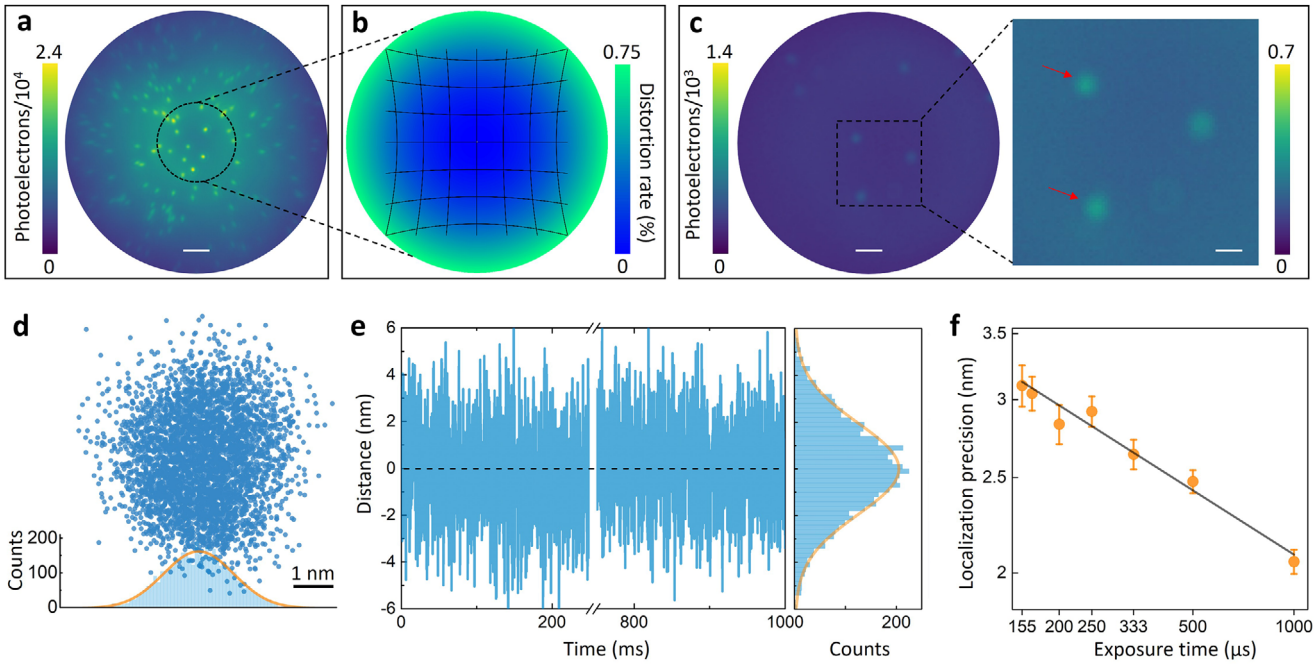

Figure 3. Determination of the localization precision under microlens chip. a) Photoelectron images of 100 nm AuNPs in the FOV imaged with microlens chip. Photoelectron data were acquired at an exposure time of 15 ms with white light illumination of 0.5 W cm^{-2}. Scale bar = 2 μm. b) Calculation results of distortion rate in the corresponding area (black dashed circle) from (a). c) Photoelectron images of immobilized 60 nm AuNPs. The red arrows indicate the adjacent AuNPs used to assess the localization precision. Scale bar = 2 μm (left image) and 1 μm (right image), respectively. d) Scatter plots of a 60 nm immobilized AuNP imaged at 200 μs exposure time with white light illumination of 1 W cm^{-2}. The lower image shows the histogram of its x coordinates. e) Distance fluctuations of the adjacent AuNPs from the analysis of the image sequences in (c), n = 5 000 frames, SD = 2.05 nm. The distance fluctuations were obtained by subtracting the mean distance value from the actual distance value. Histogram (right) shows the distribution of fluctuations. f) Experimental localization precision (orange; mean ± SD, n = 3 trials per exposure) of the 60 nm AuNPs and theoretical shot-noise limited behavior (black line) measured by varying the exposure time with white light illumination of 1 W cm^{-2}.

nanoparticles located within ROI were subjected to analysis. Note that numerous microlenses ($n \approx 20$ in a dense hexagonal arrangement) are present in the FOV under a 40× objective lens, yielding a cumulative ROI of up to ≈750 μm^2. For specific applications that require large-scale observation area, larger microlenses and post-image processing based on distortion correction can be used to further extend the usable range of ROI.[37]

In our experiments, to avoid mistaking sample vibration-induced localization errors, we evaluate localization precision by quantifying immobilized interparticle distances and express it in terms of the standard deviation (SD) of the measured distances (Figure 3c). An example of scatter plots of an individual 60 nm immobilized AuNP imaged at 200 μs exposure time is depicted in Figure 3d. We show that a single particle localization precision of 2.9 nm ($\sqrt{2} \times$ SD, SD = 2.05 nm) at 200 μs exposure time under illumination power of 1 W cm^{-2} can be achieved (Figure 3e). This is on par with state-of-the-art nanoparticle localization techniques on such short timescales, but now this is achieved with a much simpler optical configuration at significantly lower illumination power (>1000-fold). The plot of localization precision versus number of photoelectrons at different exposure times further confirms the relationship between the shot noise limited and experimentally determined localization precision (Figure 3f).

Having captured immobilized AuNPs with high spatiotemporal resolution using the microlens chip, we tracked moving AuNPs in suspension. AuNPs dispersed in water (~5 × 10^9 particle/mL) were dropped on the chip. As shown previously, under the virtual imaging mode, focal plane of the microlens, corresponding to its working distance $WD_{\mu lens}$, can be adjusted by altering the distance $D_{\mu\text{-}O}$ between the microlens and the objective, thereby enabling the observation of the objects at different z-axis positions (**Figure 4a**). To observe the moving AuNPs, we adjusted the $WD_{\mu lens}$ upward by reducing the distance $D_{\mu\text{-}O}$. Note that due to the hydrodynamic drag induced wall effect, nanoparticle velocities may vary depending on their distance (h) from the plane wall (i.e., surface of the microlens). As the distance-to-diameter ratio (h/d) exceeds the critical value of 12, the wall effect demonstrates negligible impact.[39] Therefore, depending on specific investigation, the microlens' focal plane should be determined so as to target appropriate layers of nanoparticles. Calculations based on FEM and ray optics were performed to more rigorously determine how distance $D_{\mu\text{-}O}$ affects the $WD_{\mu lens}$ of the microlens (see Supporting Information). It is shown that the $WD_{\mu lens}$ can gradually elevate as the objective approaches the microlens. E.g., when the objective moves 1 μm closer to the microlens, the microlens' focal plane rises by ≈150 nm, corresponding to a conversion ratio (R_{conv}) of 0.15 (150 nm μm^{-1}), as illustrated in Figure 4b. Calculation results also show that a lower conversion ratio, i.e., <0.1, can be achieved by using smaller microlenses (Figure S8, Supporting Information). It is noteworthy that the exponential decrease in performance is a direct consequence of the rising focal plane, which leads to a significant

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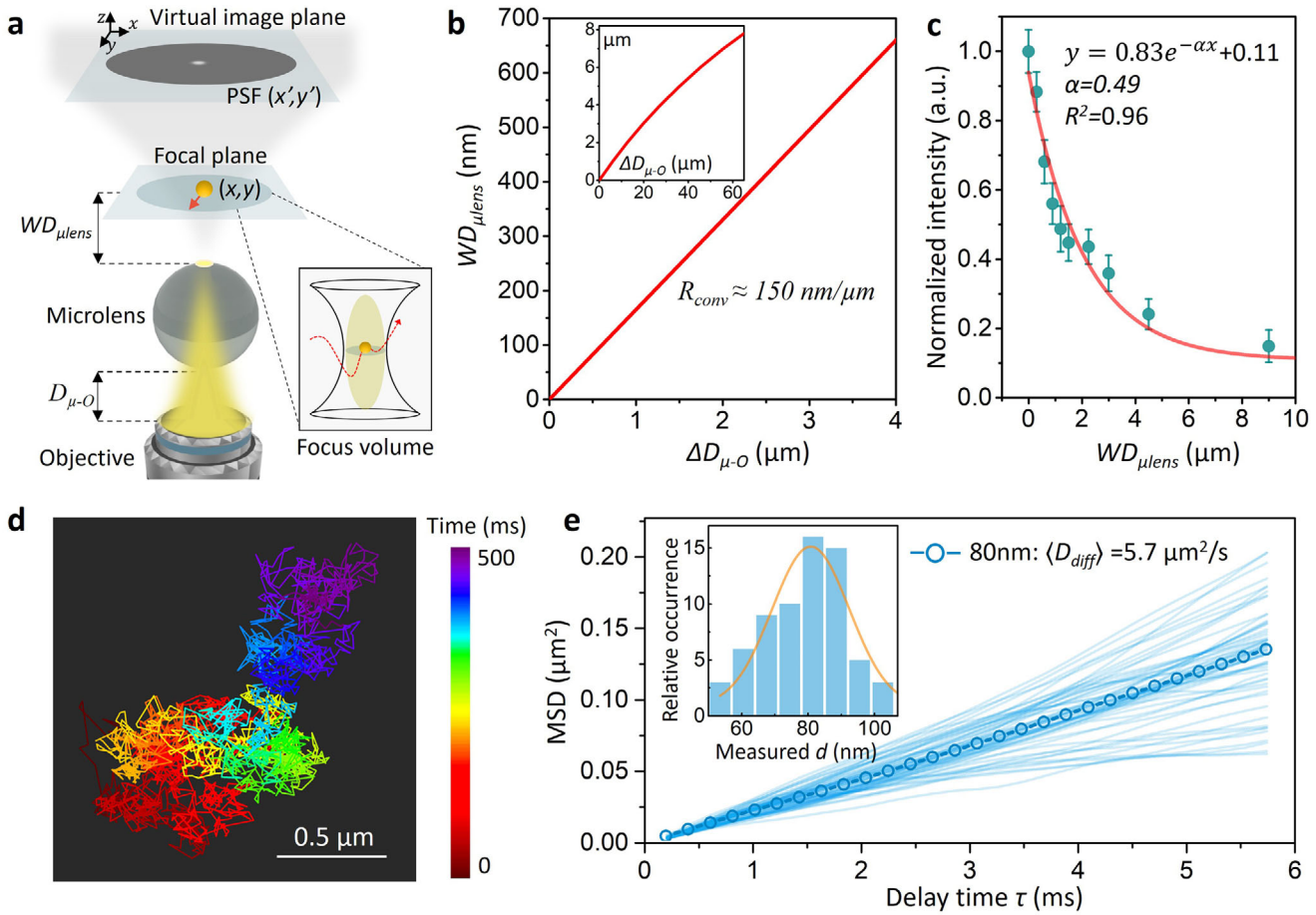

Figure 4. Tracking the moving nanoparticles in the solution. a) Illustration of the formation of the virtual image. The microlens acts as an auxiliary lens that forms a magnified virtual image above the specimen's surface that is captured by the objective lens. b) The calculated relationship between the displacement of the objective ($\Delta D_{\mu\text{-}O}$) and the working distance of the microlens ($WD_{\mu lens}$). The diameter of the microlens is 73.5 μm and the working distance of the objective is 2.3 mm as indicated by the manufacturer. Insert: the calculated $WD_{\mu lens}$ curve under the $\Delta D_{\mu\text{-}O}$ at the range of 0–65 μm. c) Normalized intensity of the AuNPs versus the working distance of the microlens. An exponential fit gives the decay constant $\alpha = 0.49$ with a squared correlation coefficient (R^2) of 0.96. d) Trajectory of a single AuNP undergoing Brownian motion recorded over 500 ms at 200 μs exposure time. e) MSD(τ) curves extracted from individual trajectories of 80 nm AuNPs (n > 50 trajectories). The thin lines show MSD data from individual trajectories that contained at least 28 trajectory points. Thick lines with circles show their weighted average (shown as mean ± SD). Diffusion constant extracted from the fits is listed in the legend. Insert: histogram of AuNP diameters extracted from the SE relation.

reduction in the intensity and SNR of the targeted AuNPs. This effect is primarily driven by the decreased ability of the microlens to efficiently collect the backscattered photons.[28,40] Experimental measurements showed a nearly exponential decay of the AuNPs intensity with the $WD_{\mu lens}$ (Figure 4c). We define the characteristic working distance WD_c as $WD_c = 1/\alpha$, where α represents the decay constant obtained from the exponential fitting in Figure 4c. For our settings, we obtain a much slow decay constant ($\alpha = 0.00049\ \text{nm}^{-1}$) thus a long decay length, which results in a WD_c of ≈2.0 μm. Compared to the commonly used TIR-based near-field imaging approaches, which typically have a characteristic working distance of ≈200 nm (limited by the penetration depth of evanescent wave), our approach offers a working range that is ten times larger, enabling the possibilities for investigations beyond surface studies. Note that owning to the highly increased photon abundance, $WD_{\mu lens}$ up to 6 μm can still be applicable without losing the advantage of high SNR ($SNR > 20$). With increased working distance, there is minimal variation in distortion rate (Figure S3, Supporting Information). Nevertheless, the induced position error can be ignored.

We are able to track moving AuNPs as small as 60 nm with high spatiotemporal resolution with the microlens chip. To further test the feasibility of our method, we track AuNPs undergoing Brownian motion and deduce their diffusivity as well as their sizes from their trajectories. In our experiments, we adjust the focal plane with enough distance ($h/d > 12$) from the substrate to diminish the wall effect as well as to ensure sufficient SNR. The diffusion of a particle in a fluid is described by the Stokes–Einstein (SE) equation, and one can arrive at its hydrodynamic diameter (d_{hydr}) by evaluating its diffusivity (D_{diff}) from the mean squared displacement (MSD) of a particle trajectory (details can be found in Supporting Information). To measure diffusivity D_{diff} with higher precision, more trajectory points and thus fast recordings are highly desirable. However, high-speed

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imaging requires a large SNR to ensure low localization error. The ability to maintain a high SNR even during high-speed imaging makes our approach an ideal technique for precise nanoparticle tracking. As an example, 80 nm AuNPs were tracked at a 200 µs time resolution. Under a specific recording time (limited by the memory buffer of the camera and the computer), the number of captured trajectories is related to the particle concentration. Meanwhile, the number of localization points in each trajectory is limited predominantly by the axial diffusion of nanoparticles. For a particle concentration of $\sim 5 \times 10^9$ particle/mL, it is possible to record more than 20 trajectories using 1-s-long videos in the ROI of a single microlens, with each trajectory consisting of over 100 localization points. An example image sequence of seven consecutive frames is shown in Figure S9 (Supporting Information). By extracting their position in x-y plane, the trajectories of individual nanoparticles can be obtained (Figure 4d). From each trajectory, the mean squared displacement curves ($MSD(\tau)$) can be obtained (thin lines in Figure 4e). The thick curve in Figure 4e shows the resulting averaged MSD plot ($n > 50$) based on a linear fit of the MSD data, which confirms the free diffusion by its linear relationship. For the 80 nm AuNPs, an excellent agreement with the theoretical value of $\langle D_{diff} \rangle = (5.7 \pm 0.3)$ µm^2 s^{-1} and a deduced hydrodynamic diameter of $d_{hydr} = 81.6$ nm was obtained.

3. Discussion and Conclusion

In summary, we present an on-chip microlens-based strategy that significantly enhances light scattering for high-performance S-SPT under conventional bright-field microscopy. By utilizing objectives with moderate NA and remarkably low illumination power, our approach facilitates broader accessibility and versatility for various applications. The microlens chip provides an enhancement range ten times greater than that of near-field optical techniques, thereby enabling investigations beyond surface studies. This concept not only makes S-SPT feasible for any laboratory equipped with a standard bright-field microscope, but also has the potential to extend to various microscopy techniques. Future work will explore optimization of microlens parameters (e.g., size, refractive index) to further enhance scattering performance. Moreover, this technique's applicability extends beyond plasmonic particles to diverse materials including polystyrene nanoparticles, opening avenues for label-free tracking of biological and synthetic nanoscale objects. Further iterations of this approach have the potential to meet the demands of applications in biophotonics, biosensors, and diagnostics where compactness and high performance are paramount.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

H.Y., Y.Z., X.T., and P.Z. developed the concept and the methodology. P.Z. and T.Z. fabricated the microlens chip. P.Z. and G.G. performed FEM simulation. P.Z. performed the SPT experiments and analyzed the data. P.Z., C.L., T.Z., and Y.Z. performed the image analysis. P.Z., X.T., and H.Y. prepared the manuscript. All authors discussed the results.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

biosensors, light scattering enhancement, nanometer localization precision, on-chip microlens, scattering-based single-particle tracking (S-SPT)

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